Intuitive Understanding of Dot Product & Matrix Multiplication

We use a simple house investment example to explain what a dot product means, why matrix multiplication is row × column, and what the result represents.

1. Problem

Many people struggle with three questions:

- Why is matrix multiplication **row × column**?
- What does a **dot product** really mean?
- What does the **result** represent?

We will explain everything using a house investment example.

2. Represent Houses as Vectors

Suppose we have 3 houses, and each house has 4 features: location, age, area, years occupied.

$$X = \begin{vmatrix} 8 & 5 & 120 & 3 \\ 6 & 15 & 90 & 10 \\ 9 & 2 & 150 & 1 \end{vmatrix} \quad \text{(3 houses × 4 features)}$$

House	Location	Age	Area	Years Occupied
House 1	8	5	120	3

House	Location	Age	Area	Years Occupied
House 2	6	15	90	10
House 3	9	2	150	1

> **Meaning:**
> Each **row** = one house
> Each **column** = one feature

3. Introduce a "Perspective"

Different people evaluate houses differently. For example, an investor may care more about location and area, and dislike an old house.

$$w_{invest} = \begin{vmatrix} 0.8 \\ -0.5 \\ 1.2 \\ -0.1 \end{vmatrix} \quad \text{(4 features × 1 perspective)}$$

This means:

- location → positive
- age → negative
- area → positive
- years occupied → slightly negative

> **This vector is a scoring rule.**

4. Dot Product (Core Idea)

Now compute House 1 · w_{invest}:

$$\begin{vmatrix} 8 & 5 & 120 & 3 \end{vmatrix} \cdot \begin{vmatrix} 0.8 \\ -0.5 \\ 1.2 \\ -0.1 \end{vmatrix} =$$

$$8\times0.8 + 5\times(-0.5) + 120\times1.2 + 3\times(-0.1) = \mathbf{147.6}$$

> **Meaning of dot product:**
> Use one perspective to score one object.

So here, **House · Investor perspective** means: *How valuable is this house from an investor's view?*

5. Multiple Perspectives → Matrix W

In real life, we may have multiple ways to evaluate the same house, such as investment value and living comfort.

$$W = \begin{vmatrix} 0.8 & 0.6 \\ -0.5 & -0.2 \\ 1.2 & 0.9 \\ -0.1 & 0.3 \end{vmatrix} \quad \text{(4 features × 2 perspectives)}$$

Each **column** is one perspective:

- Column 1 → investment value
- Column 2 → living comfort

6. Matrix Multiplication: X @ W

Now we compute:

$$Y = \begin{vmatrix} 8 & 5 & 120 & 3 \\ 6 & 15 & 90 & 10 \\ 9 & 2 & 150 & 1 \end{vmatrix} @ \begin{vmatrix} 0.8 & 0.6 \\ -0.5 & -0.2 \\ 1.2 & 0.9 \\ -0.1 & 0.3 \end{vmatrix} = \begin{vmatrix} 147.6 & 112.7 \\ 104.3 & 84.6 \\ 186.1 & 140.3 \end{vmatrix}$$

(3 houses × 2 scores)

House	Investment Value	Living Comfort
House 1	147.6	112.7
House 2	104.3	84.6
House 3	186.1	140.3

7. Key Insight

> **Matrix multiplication = a batch of dot products.**

$$Y[i][j] = \begin{vmatrix} x_{i1} & x_{i2} & \cdots & x_{in} \end{vmatrix} \cdot \begin{vmatrix} w_{1j} \\ w_{2j} \\ \vdots \\ w_{nj} \end{vmatrix} = \text{house}_i \cdot \text{perspective}_j$$

So:

- row → house
- column → perspective

- value → score

8. Why "Row × Column"?

Because we are doing this: a house (row) vs a perspective (column).

Each number in the result is one dot product between one row from **X** and one column from **W**:

$$\begin{vmatrix} 8 & 5 & 120 & 3 \\ 6 & 15 & 90 & 10 \\ 9 & 2 & 150 & 1 \end{vmatrix} @ \begin{vmatrix} 0.8 & 0.6 \\ -0.5 & -0.2 \\ 1.2 & 0.9 \\ -0.1 & 0.3 \end{vmatrix} \rightarrow \; Y[2][2] = 84.6$$

> So row × column is not arbitrary. It naturally matches:
> **object × evaluation rule**.

9. Unified View

- **Embedding (X)** → what you are
- **Weights (W)** → how I evaluate you
- **Dot product** → how valuable you are under this view

$$\text{objects} \begin{vmatrix} 8 & 5 & 120 & 3 \\ 6 & 15 & 90 & 10 \\ 9 & 2 & 150 & 1 \end{vmatrix} @ \begin{vmatrix} 0.8 & 0.6 \\ -0.5 & -0.2 \\ 1.2 & 0.9 \\ -0.1 & 0.3 \end{vmatrix} \text{perspectives} \;=$$

$$\begin{vmatrix} 147.6 & 112.7 \\ 104.3 & 84.6 \end{vmatrix} \text{scores}$$

```
186.1  140.3
```

> **Matrix multiplication = evaluate many objects under many perspectives.**